Error Propagation in Cascaded Speech Translation under Acoustic Channel Degradations: A Black-Box Case Study of Commercial Speech Intelligence APIs

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Abstract

Cloud speech services are consumed as black boxes, yet the acoustic channels feeding them are rarely clean. We present CascadeBench, a fully seeded black-box benchmark that measures how twelve controlled acoustic channel degradations propagate through the recognition and translation stages of commercial speech APIs: five recognition systems spanning three product classes of a single provider (specialised streaming, audio-LLM, and omni-modal), two cascaded and two end-to-end speech translation paths, over a symmetric 480-sentence Mandarin–English corpus with a babble severity sweep from 20 to −10 dB. Recognition error rises as a knee-shaped function of noise level: the audio-LLM systems hold a flat sub-percent floor and tolerate 2.5 to 4 dB more noise than the specialised streaming system before error reaches ten percent, but collapse sharply below 0 dB, where insertion-heavy hallucination drives character error rates above 100 percent. A telephone-band specialist shows no in-band advantage and is uniquely fragile under reverberation. Cascade translation quality falls linearly with recognition error rate (0.44 to 0.54 BLEU points per percentage point, sentence-level bootstrap 95% confidence intervals within ±0.04, intercept indistinguishable from zero) across fifteen operating conditions. End-to-end translation is not automatically more robust: under 0 dB babble the omni-modal path loses a third of its BLEU score, more than a well-frontended cascade, whereas a streaming end-to-end service is the most resilient path measured. A dual-reference protocol exposes a machine-reference scoring bias that

can invert system rankings. The complete recipe, corpus, degraded audio, and 33,000 logged API responses are released for re-measurement as services evolve.
Keywords: speech recognition, speech translation, robustness benchmark, acoustic channel degradations, cascaded systems, cloud APIs

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1. Introduction

Speech has become a primary interface to cloud intelligence. Voice assistants, meeting transcription products, live captioning services, and multilingual communication platforms are routinely assembled from commercial automatic speech recognition (ASR) and machine translation (MT) interfaces offered by large cloud providers. In a typical deployment, audio captured by a far-field microphone or transported over a communication channel is first converted to text by an ASR service, and the resulting text is then passed to a text translation engine whenever the application crosses a language boundary. This cascade architecture dominates industrial practice because it allows each stage to be selected, replaced, and billed independently, and because the best available components at each stage are often operated by different providers. An alternative architecture, end-to-end speech translation (E2E ST), maps source-language audio directly to target-language text and has matured rapidly with large foundation models.

The acoustic pathway between a speaker and these services is rarely pristine. Speech arrives mixed with competing talkers, shaped by room reverberation, band-limited by telephone channels, compressed by low-bitrate codecs, or corrupted by packet loss on unstable network links. The robustness of ASR systems to such degradations has been a central concern of the speech research community for decades, and controlled robustness corpora have driven substantial progress. What has changed is the identity of the systems that now dominate deployment. Commercial speech services are increasingly built on large language model (LLM) based ASR, on general-purpose multimodal foundation models that accept audio alongside text, and on streaming speech translation

models, none of which can be fine-tuned, inspected, or benchmarked offline by their consumers. A systems integrator who composes such services into a translation pipeline has no principled way to predict how recognition errors induced by an acoustic channel will propagate into the final translation, or whether a cascade or an end-to-end design is the safer choice under the conditions of a given deployment.

The evaluation gap is not academic. Deployment decisions are made weekly by integrators who must choose among service tiers and architectural patterns with nothing but clean-audio demo pages and marketing claims, while the operating conditions of real audio pathways, conference rooms, vehicles, telephone trunks, and congested wireless links, are precisely the conditions that clean demos do not exercise. A deployment that selects a recognition frontend on clean-audio accuracy alone can be quietly losing most of its translation quality in the field, and a deployment that assumes end-to-end robustness can be more fragile than the cascade it replaced. Closing this gap requires measurement that is controlled at the acoustic layer, complete at the pipeline layer, and reproducible at the service layer, and that is the design brief of the benchmark presented here.

This gap motivates the present study. Academic robustness benchmarks such as Aurora [1] and the CHiME challenge series [2] evaluate open recognizers on fixed noise corpora, but they do not cover the black-box commercial systems that now carry production traffic, they rarely extend beyond recognition to the downstream translation stage, and they cannot be re-run on new model versions without access to model internals. Conversely, studies that compare commercial ASR services report error rates on fixed test sets but typically consider clean audio and stop at the transcription boundary. The propagation of acoustic degradation through a complete speech translation pipeline, comparing architectural families and both cascade and end-to-end designs under identical and fully controlled channel conditions, has to our knowledge not been quantified for the current generation of speech APIs.

This paper presents CascadeBench, a black-box benchmark designed to an-

swer four questions. First, how sensitive are the main product classes of commercial ASR, namely specialised streaming models, LLM-based recognisers, and general-purpose multimodal models, to eight canonical acoustic channel degradations? Second, how does recognition accuracy scale with noise severity across the full range from mild to destructive, and where does each class fail? Third, when recognition errors flow into a downstream translation engine, does the cascade amplify them, and how does the cascade compare with end-to-end speech translation under identical conditions? Fourth, what are the latency consequences of each choice? To answer these questions under the constraint that only public interfaces are available, the benchmark synthesises a stratified bilingual corpus of 480 Mandarin Chinese and English sentences with multiple neural voices, applies a grid of eight controlled acoustic degradations together with an eight-level noise severity sweep from 20 to -10 dB, and measures five recognition systems, three cascade translation paths, and two end-to-end speech translation paths through their public APIs, using only free service quotas. All systems come from one provider, so the study is a single-provider case study whose class-level findings require cross-provider replication before attribution to architecture; this constraint is examined in Section 6 and revisited throughout.

The study yields findings that are relevant to both the science of robust speech processing and the practice of assembling speech services. Recognition services from the three classes differ sharply in where, rather than how fast, they fail under noise: the LLM-based recognisers hold a sub-percent error floor and cross ten percent error at noise levels 2.5 to 4 dB lower than the specialised streaming system, with sharper collapse beyond their knee, and below 0 dB they fail by insertion-heavy hallucination that drives character error rates above 100 percent. The end-to-end translation paths, often assumed to be more robust than cascades because they avoid error propagation, do not inherit robustness automatically: the omni-modal path loses more of its clean quality under severe noise than a well-frontend cascade, while a streaming end-to-end service is the most resilient path measured. Across fifteen operating conditions, the BLEU score of every cascade measured falls linearly with its recognition error rate,

with a pass-through slope of 0.44 to 0.54 BLEU points per percentage point, sentence-level bootstrap confidence intervals within ± 0.04, and an intercept statistically indistinguishable from zero, which gives integrators a calibrated engineering rule for estimating translation losses from recognition quality alone. Finally, the benchmark surfaces behavioural differences that are invisible in clean-audio evaluations, including two distinct empty-output mechanisms under severe noise, an explicit protocol-level refusal and a silent empty return, and the unique fragility of a telephone-band specialist under reverberation.

The contribution of this paper is threefold. It provides, to our knowledge, the first systematic measurement of acoustic degradation propagation across recognition and translation stages for commercial black-box speech APIs, covering three product classes of one provider and both cascade and end-to-end translation designs, which we frame as a case study pending cross-provider replication. It introduces a fully reproducible benchmark recipe in which the test material, the degradations, and every model response are controlled, logged, and released, so that the measurements can be repeated as providers update their models. And it distils the measurements into quantitative design guidance, including noise-failure thresholds per product class, the error pass-through regression for cascades, and a dual-reference protocol that exposes and controls the systematic bias that arises when automatic translation quality is scored against references produced by models related to one of the systems under test.

The remainder of the paper is organised as follows. Section 2 reviews related work on robust ASR, cascade and end-to-end speech translation, and the evaluation of commercial services. Section 3 describes the benchmark design, including the corpus, the degradation grid, the systems under test, and the evaluation protocol. Section 4 presents the results, and Section 5 discusses their implications for practitioners and for future evaluation practice. Limitations are examined in Section 6, and Section 7 concludes.

2. Related work

2.1. Robustness of speech recognition under acoustic degradation

The sensitivity of ASR to noise and channel distortions has been studied since the earliest statistical recognisers [3]. Surveys such as that of Virtanen et al. [4] organise this literature around feature enhancement, model adaptation, and multi-condition training. Controlled evaluation infrastructure has been decisive for progress. The Aurora framework [1] standardised noise types and signal-to-noise ratios for telephone-band recognition, and the CHiME challenge series [2] extended such evaluation to far-field, reverberant, and overlapping-speech conditions. Classical feature transformations such as RASTA filtering [5] targeted the same distortions from the signal-processing side, and synthetic room impulse responses generated with the image method of Allen and Berkley [6] and its modern implementations [7] remain standard tools for controlled reverberation studies. More recently, data augmentation strategies such as SpecAugment [8] have embedded artificial degradation into the training process itself, and weakly supervised training at scale, as exemplified by Whisper [9], has produced recognisers with markedly improved noise tolerance. The present work inherits the methodological principle of these efforts, namely controlled and reproducible degradation of the input signal, but applies it to black-box services rather than to trainable models, which precludes training-based countermeasures and makes the measurement itself the object of study.

2.2. Cascaded and end-to-end speech translation

The cascade versus end-to-end question has accompanied speech translation since neural systems became practical. Early sequence-to-sequence systems demonstrated that audio-to-text translation without an explicit recognition stage is feasible [10, 11], and Bentivogli et al. [12] compared the two paradigms systematically on IWSLT data, finding that end-to-end systems could match cascades in translation quality while differing in error structure, with cascades inheriting recognition errors and end-to-end systems producing more

fluent but occasionally freer output. The survey of Sperber and Paulik [13] synthesises this literature, including work on robustness of end-to-end systems to noisy input. Subsequent research addressed noise robustness of end-to-end ST through self-training and data selection [14]. These studies evaluate models that the researchers train themselves on public corpora. Robustness of the end-to-end component in particular has been treated as a training-data problem, with noise-augmented source audio, self-training on noisy transcripts, and domain adaptation as the standard remedies. That framing presumes control over the model, which is exactly what a service consumer lacks. When the translation component is a fixed API, the robustness question shifts from how to train the model to how to select among available models and architectures under measured channel conditions, and that selection question is what the present benchmark operationalises. The commercially relevant question, namely how the two architectures behave when every stage is a fixed, provider-operated service selected and combined by an integrator, has received far less attention, and the present benchmark is designed specifically for that setting.

2.3. Evaluation of commercial speech services

A third strand of related work evaluates speech services as they are exposed to users. Benchmarks for large audio-language models, such as AIR-Bench [15] for audio understanding and VoiceBench [16] for LLM-based voice assistants, measure knowledge, comprehension, and instruction following, and they establish the black-box, generative-evaluation methodology that the present benchmark extends to the acoustic-robustness axis; neither, however, characterises behaviour under controlled channel degradation, and both stop at the response boundary rather than following errors into downstream tasks. The propagation of recognition errors into translation has a long history in the cascade literature, going back to the statistical framework of Ruiz and Federico [17] for relating ASR error types to downstream MT quality, but that line of work assumes access to the models and corpora involved rather than opaque commercial endpoints. Methodologically, black-box evaluations face constraints that mirror

ours: no access to model internals, no control over model versions, and reliance on text normalisation to compare outputs against references. They have, however, largely concentrated on clean or moderately degraded audio, and on transcription accuracy alone. The broader evaluation culture in speech and audio quality research, with its calibrated degradation methodology and emphasis on reproducibility, similarly targets the acoustic layer rather than the linguistic stages downstream, and it is that tradition which the present service-level benchmark follows, adding the service-level measurement that intelligibility metrics cannot substitute for, as the dissociation documented in Section 4.3 shows. The contribution of the present study relative to this strand is twofold. First, the evaluation is extended across the translation boundary, quantifying how recognition differences propagate into translation quality for both cascade and end-to-end architectures. Second, the acoustic degradations are applied under full experimental control, with calibrated noise levels, a synthetic room impulse response, codec and packet-loss emulation, and fixed random seeds, which makes the entire measurement reproducible rather than dependent on whatever happened to be in the test recordings.

2.4. Positioning

In summary, robustness benchmarks address open models but not commercial services, service evaluations address commercial ASR but usually only in clean conditions and only at the recognition boundary, and the cascade-versus-end-to-end literature addresses research models rather than provider-operated systems. CascadeBench combines the three perspectives. It applies the controlled degradation methodology of the robustness literature to the black-box commercial setting, extends the measurement through the translation stage for both architectures, and adds a dual-reference scoring protocol that quantifies a bias source that any evaluation involving modern generative reference translations must confront. A systematic search of the speech-translation and audio-benchmarking literature, including the benchmarks cited above, found no previous study that measures all of these dimensions simultaneously for the

current generation of speech APIs, and none that reports the linear recognition-to-translation error pass-through regression that Section 4 documents; we state this as a finding of our search rather than as a proof of absence.

3. Materials and methods

3.1. Design overview

CascadeBench measures how acoustic channel degradations propagate through commercial speech recognition and translation services that can only be accessed through their public interfaces. The design is shown in Fig. 1. A bilingual corpus of 480 sentences is synthesised to audio with multiple neural voices, degraded by a grid of controlled channel impairments, and submitted to five recognition systems and four speech translation paths. Every stage of the pipeline is scripted with fixed random seeds, and every model response together with its round-trip latency is logged at the sentence level, so that the complete measurement can be re-executed as providers update their services.

3.2. Corpus

The corpus contains 240 Mandarin Chinese and 240 English sentences, balanced across four content domains and three length strata, twenty sentences per cell. The domains are broadcast news, everyday conversation, technical and scientific exposition, and a numerals-and-entities stratum rich in dates, quantities, addresses, and named entities, which is known to stress recognition and inverse text normalisation. Short sentences contain 8 to 14 Chinese characters or 5 to 9 English words, medium sentences 15 to 25 characters or 10 to 17 words, and long sentences 26 to 39 characters or 18 to 24 words. The mean length is 19.3 characters for Chinese and 12.0 words for English. Sentences were authored to be self-contained and factually neutral, to avoid brand names and transliterated loanwords whose recognition would depend on provider-specific vocabulary rather than on acoustic processing, and to distribute numerical expressions across date, time, quantity, price, and identifier formats within the

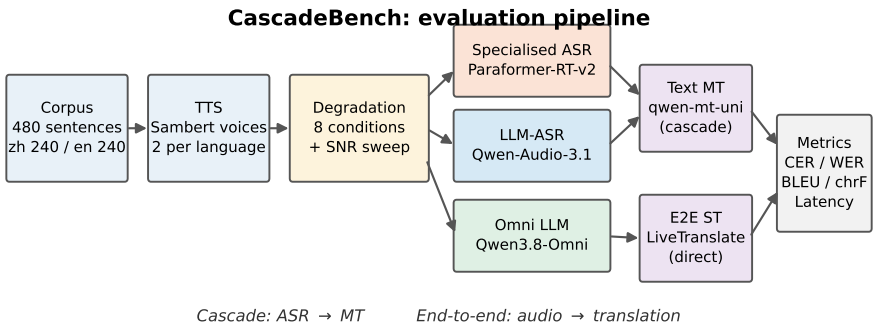


Figure 1: Architecture of the CascadeBench evaluation pipeline. A stratified bilingual corpus is synthesised with multiple neural voices, degraded by a grid of eight acoustic channel conditions plus a noise severity sweep, and processed by recognition systems from three architectural families. Cascade systems forward recogniser output to a text translation engine; end-to-end systems translate audio directly.

numerals domain. All numbers are written out in words in the corpus text, for example two thousand twenty five, so that the spoken realisation is unambiguous and the reference transcript matches the synthesis input exactly.

Each sentence was synthesised by a neural speech synthesis service (Sambert, Alibaba Cloud Model Studio) using two distinct voices per language, a male and a female voice for Chinese and two native-sounding voices for English, with sentences alternating between the voices so that each voice realises 120 sentences per language. Synthesis produced 16 kHz single-channel PCM audio. The 40 babble noise source tracks described below were synthesised with six additional voices that are disjoint from the four test voices. Using synthesis rather than recorded speech has two methodological advantages. The reference transcript is exact by construction, which removes the transcription noise that limits error floor measurement in recorded corpora, and the acoustic properties of the test material can be regenerated at any time, which decouples the benchmark from the licensing and availability constraints of recorded corpora such as AISHELL-

1 [18] or LibriSpeech-derived material [19]. The corresponding limitation, that synthetic speech does not sample the full variability of spontaneous human speech, is examined in Section 6.

3.3. Acoustic degradation grid

Eight channel conditions plus the clean control form the main grid. The conditions were selected to span the canonical impairment families that speech services encounter between a talker and a data centre.

Additive babble noise is the most extensively sampled family. Babble was constructed by concatenating randomly selected segments from the 40 disjoint noise-source tracks described above, mixing three such streams, and normalising the mixture, which produces speech-shaped noise with realistic spectral and temporal structure rather than stationary synthetic noise. Speech-shaped interference was chosen over stationary noise because it is the harder and the more ecologically valid case: babble exhibits the temporal modulation and spectral occupancy of competing talkers, masking the target in the same time-frequency regions that carry linguistic information, whereas stationary noise is partly countered by any spectral-subtraction-style front-end that a provider may have deployed internally. The noise was scaled against the active-speech level of each utterance, measured as the mean energy of frames above the 60th energy percentile, to hit target signal-to-noise ratios of 20, 15, 10, 5, 2.5, 0, -5, and -10 dB. Calibration was verified by re-measuring the realised signal-to-noise ratio of the mixed signals against the clean signal; the realised values matched the active-level targets within 0.1 dB on average over the 20 to 0 dB ladder and within 0.6 dB at the -10 dB point. The severity convention deserves emphasis because it is not the only possible one: because the 60th-percentile active-level rule ignores inter-speech pauses, the total-power signal-to-noise ratio of the same signals runs 1.8 to 2.2 dB below the stated active-level values, so a reader comparing against total-power conventions should shift our levels by approximately -2 dB. The 10 dB and 0 dB points enter the main grid, and the full eight-point ladder forms the severity sweep on a stratified 144-sentence

subset, 72 sentences per language, sampled evenly from every domain-by-length cell. For the telephone-band recogniser, the degraded signals are generated at 16 kHz first and then downsampled to 8 kHz, so the specialist sees the same noise realisation band-limited to its native input bandwidth rather than noise generated inside its band.

Reverberation was applied by convolving each signal with a synthetic room impulse response of the exponentially decaying noise form [6, 7] with a direct-path impulse and a reverberation time of 0.5 s. We measured the acoustic properties of the realised impulse responses: Schroeder T_{60} is 0.51 s as designed, the clarity ratio C_{50} is +4.9 dB, and the direct-to-reverberant ratio is −27.6 dB, the last reflecting the exponentially decaying diffuse tail of this synthesis, which concentrates reverberant energy in a spectrally flat late field rather than in discrete early reflections. The condition is therefore best read as diffuse late reverberation at a moderate clarity ratio rather than as an emulation of a specific small room, and its contrast with real recorded impulse responses, whose early-reflection structure colours the spectrum more strongly, is examined in Section 6. Narrowband degradation resamples each signal to 8 kHz and back to 16 kHz with polyphase anti-aliasing filters, which removes the 4 to 8 kHz band exactly as a telephone channel does; the telephone-specialised recogniser receives the 8 kHz signal natively. Codec degradation encodes each signal with the Opus codec [20] at 8 kbit/s mono, in the default voip application mode of the libopus encoder as shipped by ffmpeg 7.1.5, and decodes it back. Measurement of the decoded signals shows that at this bitrate the encoder operates effectively as a narrowband coder: the decoded output retains less than 0.1 percent of clean high-band energy above 4 kHz, so the codec and narrowband conditions are not fully independent stressors of high-frequency content. Packet loss zeroes 15 percent of 20 ms frames chosen uniformly at random, emulating lossy transport without jitter compensation. Clipping applies hard amplitude limiting at 0.3 of full scale, which with a mean input peak of 0.6 clips about 0.6 percent of samples. Tempo distortion resamples the time axis by a factor of 1.15 with pitch preservation using the ffmpeg atempo filter; the spectral centroid is preserved

within one percent, no level renormalisation is applied after any degradation, and the peak-normalised output convention follows the synthesis. All random draws use per-sentence and per-condition seeds derived deterministically from a CRC32 digest of the sentence identifier and condition name, and the released audio tree, generated with the same rule, is shipped alongside the logs so that the original measurements are exactly reproducible while fresh regenerations remain deterministic.

Formally, for a clean signal $x[n]$ of length N and a babble realisation $b[n]$, the mixed signal at level ρ dB is

$$y[n] = x[n] + \sigma\, b[n], \qquad \sigma = \sqrt{\frac{P_{\mathrm{act}}(x)}{10^{\rho/10}\,\overline{P}(b)}}, \tag{1}$$

where $\overline{P}(b)$ is the mean power of the babble and $P_{\mathrm{act}}(x)$ is the active-speech power estimated from frames whose energy exceeds the 60th energy percentile of the utterance, which approximates the ITU-T P.56 active-level convention with a fixed percentile rule. The reverberation impulse response is

$$h[n] = \delta[n] + v[n]\, e^{-6.9\, n/(f_s T_{60})}, \qquad n = 0, \ldots, \lceil f_s T_{60} \rceil, \tag{2}$$

with $v[n]$ unit-variance white noise, $f_s = 16$ kHz, $T_{60} = 0.5$ s, and h normalised to unit energy, and the reverberant signal is the truncated convolution of x with h rescaled to the peak amplitude of x. Packet loss zeroes each independent 20 ms frame with probability $p = 0.15$,

$$y[n] = m_k\, x[n], \quad n \in \text{frame } k, \qquad m_k \sim \text{Bernoulli}(1-p), \tag{3}$$

and clipping applies $y[n] = \mathrm{sign}(x[n])\min(|x[n]|, 0.3)$. Narrowbanding resamples by factors of one half and two with polyphase anti-aliasing filters, codec degradation round-trips the signal through the Opus encoder at 8 kbit/s, and tempo distortion applies a pitch-preserving time-scale modification of factor 1.15.

3.4. Systems under test

Table 1 lists the systems. The recognition systems span three architectural families as served by a single commercial provider (Alibaba Cloud Model Studio,

Beijing region) during October 2026, accessed through their documented interfaces. The specialised family consists of streaming recognisers in the Paraformer lineage [21], a 16 kHz model and an 8 kHz telephone-band model. The LLM-ASR family consists of Qwen-Audio 3.1 Flash [22] and Qwen3-ASR Flash, audio input models built on large language model backbones. The multimodal family consists of Qwen3.8-Omni Flash, a general-purpose omni-modal foundation model [23] queried with a transcription instruction. All systems were accessed in their default configurations, since a consumer of a black-box service cannot apply front-end training or adaptation, and the question of interest is precisely how the services behave out of the box. The streaming recognisers were driven by streaming local audio over their realtime WebSocket interface, sending each utterance as 16 kHz or 8 kHz PCM frames at the maximum accepted rate and collecting sentence-final segments. The LLM-ASR and omni systems received base64-encoded whole utterances in their audio-input message format, the omni model together with an instruction prompt for transcription or translation. The streaming translation service was driven over its realtime interface with a session configuration specifying text-only output and the target language, sending raw PCM frames and collecting both the source transcript and the translated text that the service emits, and the text translation engine received source text with explicit source and target language tags. Every call was executed with an automatic retry policy of up to six attempts with exponential backoff on throttling, timeout, and connection failures, and no other tuning parameters were exposed or set. The speech translation systems are the text translation engine qwen-mt-uni, used as the second stage of three cascades built on the 16 kHz specialised recogniser, on the Qwen-Audio 3.1 recogniser, and on the Qwen3-ASR recogniser, and two end-to-end paths: the omni foundation model with an audio translation instruction, and a dedicated streaming speech translation service (LiveTranslate) that internally combines a streaming frontend recogniser with a translation model. The internal frontend of the streaming translation service, which its responses expose as source transcripts, is additionally scored as a sixth recognition system on the severity sweep; the five purchasable recogni-

Table 1: Systems under test and their roles in the benchmark. All systems are operated by a single provider (Alibaba Cloud Model Studio), so product class and vendor are confounded by design; the provider does not disclose the internal architectures, and the class labels follow the product positioning.

System (product class)	Interface	Role	Evaluated on
Paraformer-RT-v2 (specialised)	streaming ASR	recogniser	full grid
Paraformer-RT-8K-v2 (specialised, 8 kHz)	streaming ASR	recogniser	Chinese subset
Qwen-Audio-3.1-Flash (LLM-ASR)	audio input API	recogniser	full grid
Qwen3-ASR-Flash (LLM-ASR)	audio input API	recogniser	full grid
Qwen3.8-Omni-Flash (omni)	audio input API	recogniser, E2E ST	main grid
LiveTranslate frontend (specialised)	streaming ST	recogniser (internal)	sweep subset
qwen-mt-uni (MT)	text translation	cascade stage, oracle	full corpus
Qwen3.8-Max (LLM)	text generation	reference translations	full corpus

tion systems of Table 1 are therefore complemented by this embedded frontend wherever it is exposed. All API calls were executed on free-tier quotas with the stop-on-exhaustion option enabled, which both bounds the study at zero monetary cost and documents the practical quota requirements of such a measurement. All systems were accessed with their default decoding configurations and no exposed sampling parameters were overridden, so the repetition analysis of Section 4.8 characterises the services as a consumer actually receives them.

3.5. Evaluation protocol

Recognition accuracy is scored as the character error rate (CER) for Chinese and the word error rate (WER) for English, computed with the standard Levenshtein alignment [24] after a normalisation pipeline that is applied identically to references and hypotheses. The normalisation performs Unicode compatibility folding, case folding, punctuation removal, and, critically, canonicalisation of numeric expressions. Because commercial recognisers differ in their inverse text normalisation conventions, with the same spoken quantity rendered as twenty six, 26, or two six depending on the system, both sides of every comparison are mapped to a canonical digit form by rules that handle Chinese and English

cardinal numbers, digit-by-digit readings, time expressions, decimal fractions, and scale multipliers such as wan and yi on the Chinese side and million and billion on the English side, with ordinal expressions masked to a common token. The rule set was developed and verified on the clean condition, where residual normalisation artefacts were inspected sentence by sentence until the clean-condition error rates reported in Section 4 reflect genuine recognition behaviour rather than scoring artefacts. Utterances for which a service returns no transcript at all, a behaviour documented in Section 4, are scored as complete failures with an error rate of one.

Translation quality is scored with corpus-level BLEU [25] using sacreBLEU [26] with the 13a tokeniser for English targets and the character-level tokeniser for Chinese targets, and with chrF [27] as a second string-matching metric; sacreBLEU signatures are fixed in the released scripts (BLEU: effective order 1, case-insensitive, 13a for English and zh character tokenisation for Chinese). Because the corpus is direction-symmetric, BLEU is computed separately for the Chinese-to-English and English-to-Chinese halves and macro-averaged, and the number of scored sentences n is stated for every cell because coverage varies where frontends refuse or sessions drop. Two scoring protocols are distinguished wherever they differ: quality conditional on a non-empty translation, and effective quality in which a refused frontend output is scored as an empty translation. Corpus-level BLEU reacts to empties only through the brevity penalty, which saturates when hypotheses are longer than references, so the refusal-sensitive comparison in Section 4.4 is anchored on chrF as well. Every hypothesis is scored against two reference sets. The primary reference is produced by Qwen3.8-Max [28], the strongest general-purpose text LLM of the same provider, translating the clean reference transcripts; it shares a vendor, though not a product line, with the translation systems under test, and the family relationships of all systems and references are disclosed in Table 1 and revisited in Section 4.5. The secondary reference is the oracle cascade output, that is, the cascade translation engine applied to the clean reference transcripts. The dual-reference design exposes a bias that Section 4.5 quantifies: automatic scoring against machine-

generated references systematically favours systems from the same model family as the reference generator, and conclusions about absolute quality depend on the choice of reference, whereas conclusions about degradation are stable. The text normalisation rules were developed by inspecting clean-condition outputs until residual mismatches ceased; a sensitivity analysis in Appendix Appendix C re-scores the main grid without numeral canonicalisation and shows that the system ordering is preserved while clean-condition gaps compress. To quantify run-to-run variability of the non-deterministic LLM services, the Qwen-Audio and Qwen3-ASR systems were re-run twice on the 0 dB babble and tempo conditions; condition-level error rates vary by at most 0.9 percentage points across the three runs, although one in five individual transcripts differs between runs (Section 4.8).

Latency is measured as the wall-clock round trip of each API call from dispatch to the complete response, recorded per call. The text translation stage was additionally benchmarked with 120 sequential timed calls on oracle sentences, because the main cascade logs do not carry per-call timing; cascade latency is therefore reported as the sum of the recognition and translation stage means, which is the quantity a consumer experiences when the two stages are invoked sequentially. Streaming systems are fed audio at the maximum rate their interface accepts, so the measurement reflects service-side processing rather than client-side throttling, and it measures whole-utterance turnaround rather than first-partial latency; streaming systems would show substantially lower first-partial latencies than the whole-utterance figures reported here, so the two interface classes should not be compared on absolute latency. All clients ran in one region close to the service endpoint; network round trips are included in all figures.

3.6. Experimental scale and cost

Table 2 reports the exact call accounting of the complete campaign, including the extension conditions of the revision round. In total 33,847 successful API responses were logged (26,238 in the original measurement and 7,609 in the

extension), alongside 3,919 failed attempts that were retried or, in 24 cases, ended as protocol-level refusals of the Qwen-Audio system under 0 dB babble. The 1,544 successful LiveTranslate sessions cover 89.4 percent of that system's 1,728 targeted evaluations; the missing sessions are websocket drops that did not recover within the retry budget, and every reported LiveTranslate figure states its n. The omni system exhausted its complimentary quota before completing its grid: its recognition stage covers the main grid on the 144-sentence sweep subset (119 of 144 sentences at the tempo condition) and its end-to-end translation path covers 83 to 84 of the 144 sweep sentences per condition, which is why omni-based comparisons are restricted accordingly and flagged wherever they appear. The total wall-clock duration of the original data collection was approximately four hours with up to five systems queried in parallel. These figures are reported so that other laboratories can estimate the feasibility of replicating or extending the measurement; they also demonstrate that a multi-system, multi-condition robustness characterisation of commercial speech services is attainable at zero monetary cost with careful quota budgeting, which lowers the barrier for the periodic re-measurement that the field currently lacks.

3.7. Statistical analysis

All aggregate error rates and quality scores are reported with 95 percent bootstrap confidence intervals over sentences, using 1000 resamples. Error rates are sentence-level: the character error rate for Chinese targets and the word error rate for English targets, each computed with the standard Levenshtein alignment, so a Chinese character rate and an English word rate are pooled only as a macro-average of the two language halves, and an insertion-dominated transcript can push a rate above 100 percent, which occurs in the sub-zero-dB regime documented in Section 4.3. Differences between systems within a condition are tested with the Wilcoxon signed-rank test [29] on paired per-sentence errors, with Holm correction [30] within the family of all pairwise comparisons of the same condition and language; because both systems are often simultaneously perfect, the fraction of zero differences is reported alongside (up to 83

Table 2: API call accounting: successful responses and unique evaluations per phase. Coverage notes: the 8 kHz specialist is evaluated on the Chinese subset; the omni system is restricted by quota exhaustion as described in the text.

Phase	System	Calls	Coverage
Synthesis	Sambert (4 test + 6 babble voices)	520	480 corpus + 40 babble
Recognition	Paraformer-RT-v2	4,752	full grid + sweep
Recognition	Paraformer-RT-8K	2,376	Chinese subset + zh sweep
Recognition	Qwen-Audio-3.1	4,752	full grid + sweep (24 refusals)
Recognition	Qwen3-ASR	4,752	full grid + sweep
Recognition	Qwen3.8-Omni	1,271	main grid on sweep subset
Recognition	LiveTranslate frontend	—	scored from LT sessions
Translation	oracle / reference	960	480 + 480
Translation	cascades (2 frontends)	3,456	12 conds × 144
Translation	E2E omni	1,004	83–84 of 144 × 12 conds
Translation	E2E LiveTranslate	1,544	89.4% of 12 × 144
Extension	4 ASR systems × 3 new SNRs	1,512	2.5, −5, −10 dB × 144
Extension	cascades on new SNRs	864	2 frontends × 3 conds × 144
Extension	Qwen3-ASR cascade (12 conds)	1,728	third cascade system
Extension	E2E LiveTranslate on new SNRs	400+	3 conds × 144, partial
Extension	repetition runs (2 systems)	1,152	2 conds × 2 repeats
Extension	MT latency benchmark	120	sequential timed calls

percent under reverberation), and the test is then driven by the minority of sentences where the systems differ. Noise severity is characterised two ways: by a four-parameter logistic fit to the eight-point curve, from which we report the signal-to-noise ratio at which fitted error crosses ten percent together with a sentence-level bootstrap confidence interval of the crossing, and by the maximum fitted slope in percentage points per dB as a measure of how sharp the knee is. Ordinary least squares slopes over sub-ranges are reported only for continuity with common practice, and the text notes how strongly they depend on the chosen range. The relationship between recognition error and translation

quality is summarised by a linear pass-through regression with explicit intercept, fitted over all fifteen operating conditions per cascade, with sentence-level bootstrap confidence intervals for slope and intercept, leave-one-condition-out refits, and a leverage check that excludes conditions with recognition error above 40 percent. Signal processing and statistics use SciPy [31], librosa [32], and pystoi for the short-time objective intelligibility analysis of Section 4.3.

4. Results and analysis

4.1. Clean baseline and the main degradation grid

Table 3 reports sentence-level error rates on the main grid for the five recognition systems, and Fig. 2 visualises the same data split by language. On clean synthetic speech the three foundation-model recognisers are close to their scoring floor, with error rates between 0.3 and 0.5 percent, while the specialised streaming recogniser operates at 0.56 percent on Chinese and 3.99 percent on English, reflecting its primarily Mandarin training. The telephone-band specialist reaches 1.86 percent on its native language. The clean-condition figures establish that the comparisons under degradation are not confounded by differences in clean accuracy of comparable magnitude.

Three regularities stand out in the grid. First, babble noise at 0 dB is by far the most destructive condition for every system, but the magnitude of the collapse separates the architectural families sharply. The specialised streaming recogniser reaches a pooled error rate of 58.5 percent and the telephone-band specialist 73.2 percent, whereas the two LLM-ASR systems remain at 25 to 28 percent and the omni model at 31.8 percent. At matched severity, the foundation-model recognisers commit roughly half the error of the specialised systems, and the difference is significant at the sentence level: in Chinese at 0 dB, Qwen-Audio 3.1 errs at 6.3 percent against 41.6 percent for the streaming specialist (Wilcoxon $p < 0.001$, $n = 240$).

Second, the effect of noise is strongly language-asymmetric in a way that differs by class. For the LLM-ASR systems, Chinese recognition at 0 dB remains

Table 3: Mean recognition error rates in percent on the main degradation grid, both languages pooled, with bootstrap 95 percent confidence intervals in parentheses. The 8 kHz specialist is evaluated on the Chinese subset only.

System	Clean	Babble 10	Babble 0	Reverb	Narrow	Codec	Loss	Clip	Tempo
Paraformer-RT-v2	2.3	2.8	58.5	3.2	2.1	2.4	3.2	2.5	2.3
	(1.7,2.9)	(2.2,3.6)	(54.5,62.6)	(2.5,4.0)	(1.5,2.7)	(1.8,3.1)	(2.5,4.0)	(1.9,3.2)	(1.7,3.0)
Paraformer-RT-8K	1.9	4.7	73.2	18.2	1.9	1.8	3.3	1.9	2.3
	(1.2,2.7)	(3.3,6.3)	(68.8,77.6)	(14.4,22.3)	(1.2,2.7)	(1.1,2.6)	(2.2,4.6)	(1.2,2.7)	(1.5,3.3)
Qwen-Audio-3.1	0.5	0.9	28.2	0.7	0.6	0.6	0.6	0.9	0.6
	(0.3,0.7)	(0.6,1.3)	(24.8,32.0)	(0.5,1.0)	(0.4,0.8)	(0.4,0.8)	(0.4,0.9)	(0.6,1.2)	(0.4,0.8)
Qwen3-ASR-Flash	0.3	0.5	25.1	0.3	0.3	0.3	0.4	0.3	0.3
	(0.2,0.5)	(0.3,0.8)	(21.7,28.7)	(0.2,0.5)	(0.2,0.5)	(0.2,0.5)	(0.2,0.6)	(0.2,0.5)	(0.2,0.5)
Qwen3.8-Omni-Flash	0.4	0.6	31.8	1.1	0.1	0.3	0.2	1.0	0.3
	(0.2,0.6)	(0.4,0.9)	(28.0,35.8)	(0.7,1.6)	(0.0,0.3)	(0.1,0.5)	(0.1,0.4)	(0.6,1.5)	(0.1,0.5)

low, at 6.3 percent for Qwen-Audio 3.1 and 13.1 percent for Qwen3 ASR, while their English error rates at the same severity reach 50.2 and 37.1 percent. The specialised streaming recogniser suffers heavily in both languages, 41.6 percent in Chinese and 75.5 percent in English. The babble mixture is language-mixed, so the asymmetry cannot be attributed to language-matched interference alone; a plausible partial explanation is that the tonal structure of Mandarin provides complementary cues that survive energetic masking, a hypothesis we state as untested since the benchmark does not isolate tone cues, and one which the general noise-robustness literature does not settle for black-box services [4]. A related measurement caveat is that Chinese is scored at character level and English at word level; the two scales are not directly comparable, and the asymmetry should be read per language rather than across languages.

Third, the non-noise degradations expose narrow specialisation rather than uniform fragility. Narrowbanding to the telephone band costs the LLM-ASR systems 0.3 to 0.6 percentage points of absolute error and the omni model 0.1, whereas the telephone-band specialist shows no measurable home-field advan-

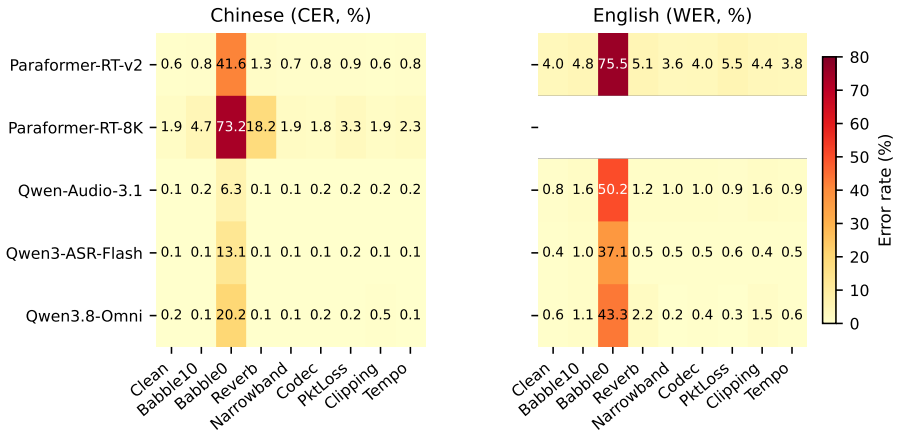


Figure 2: Recognition error rates by system and degradation condition, separated by language. Colour scale is truncated at 80 percent. The 8 kHz specialist is evaluated on the Chinese subset only.

tage over the generalist systems in its own band, where it errs at 1.9 percent against 0.1 to 0.7 percent for the generalists. Reverberation, by contrast, is uniquely expensive for the telephone-band specialist at 18.2 percent, between five and ninety times every other system's score under the same condition, which suggests that its front-end processing is tuned for direct-path telephone speech. Codec compression, packet loss, and tempo distortion at the chosen severities are individually mild for all systems, with the largest single-condition excursion being the packet-loss score of the telephone specialist at 3.3 percent against its clean 1.9. This does not make them harmless: Section 4.4 shows that even sub-point recognition changes propagate measurably into translation quality, and the conditions were deliberately calibrated to deployed-but-aggressive levels rather than to failure-inducing extremes.

Two further behaviours deserve mention because they are invisible in conventional aggregate reporting. The numeral-and-entity domain is five times more error-prone than the other domains for Qwen-Audio 3.1 on clean speech, at 1.25 against 0.25 percent, whereas Qwen3 ASR shows no such penalty, which indicates that inverse text normalisation rather than acoustics dominates that

gap; the normalisation sensitivity analysis of Appendix Appendix C supports the same reading. More strikingly, severe noise produces empty final transcripts in three of the five systems, through two distinct mechanisms. Qwen-Audio 3.1 is the only system that refuses at the protocol level, returning an explicit no-words error on 5.0 percent of its 0 dB utterances, concentrated in English at 9.2 percent of that subset; these refusals are scored as complete failures in Table 3. The specialised streaming recognisers fail silently instead: Paraformer-RT-v2 returns an empty transcript on 8.8 percent of its 0 dB utterances without any error signal, and the telephone specialist does so on 0.8 percent. Qwen3-ASR and the omni model always emit a hypothesis, however wrong. For a downstream consumer the two mechanisms differ in detectability: an explicit refusal can be caught and retried, whereas a silent empty return is indistinguishable from a genuine silent segment, and this distinction is the kind of behavioural divergence that only black-box evaluation at controlled severity can surface. The full per-system, per-condition refusal rates, including the sub-zero regime where they rise above 40 percent, are tabulated in Appendix Appendix F.

The remaining conditions merit condition-specific remarks because their effects are small in aggregate but structured in detail. Reverberation with a reverberation time of 0.5 s raises the specialised streaming recogniser to 3.2 percent pooled, but the effect concentrates in English, where it reaches 5.1 percent against 1.3 percent in Chinese, and the paired difference against the LLM-ASR systems remains significant in both languages (Wilcoxon $p < 0.01$ in Chinese, $p < 0.001$ in English; 72 to 83 percent of paired differences are zero under reverberation, so the significance is carried by the minority of sentences where the systems differ). The omni model shows an idiosyncratic signature under reverberation and clipping, at 1.1 and 1.0 percent, roughly triple its clean error rate, which suggests that its audio encoder is more sensitive to spectral colouration and non-linear distortion than the dedicated recognisers even though it holds up well under noise. Codec compression at 8 kbit/s leaves every system within 0.3 percentage points of its clean score, which is expected in retrospect because measurement of the decoded audio shows the 8 kbit/s Opus stream to be ef-

fectively narrowband, so the codec condition largely re-tests the narrowband condition at a different distortion profile. Packet loss at 15 percent and clipping at the 0.3 threshold behave as mild aggregate stressors, and tempo distortion at 1.15 times is fully transparent to every system, which is expected because time-scale modification preserves the spectral envelope that recognisers rely on. The practical reading of these patterns is that a service integrator who cannot control the acoustic environment should prioritise noise mitigation far above codec or transport tuning, since the latter contribute almost nothing to the error budget at realistic operating points.

4.2. Stratified analysis by domain

Because the corpus is balanced by construction, error rates can be decomposed by content domain without re-weighting. On clean audio the technical domain is the most error-prone for the specialised streaming recogniser at 3.3 percent, consistent with its terminological density, while the conversation domain is the easiest for every system, below 0.1 percent for the Qwen-Audio recogniser. The numerals-and-entities domain shows the clearest family split on clean speech: the two LLM-ASR systems differ by an order of magnitude on it, at 1.2 percent for Qwen-Audio against 0.15 percent for Qwen3-ASR, which as noted in Section 4.1 points to inverse text normalisation rather than acoustics as the separating factor. Under 0 dB noise the domain differences compress for every system: the spread across domains narrows to about nine points for the Qwen-Audio recogniser and to twelve points for the specialised streaming recogniser, and the ordering changes, with the technical domain becoming the most difficult for both. The compression indicates that under severe energetic masking the bottleneck shifts from lexical and normalisation knowledge, which distinguishes systems on clean speech, to the acoustic recovery problem, which is common to all systems and dominates the error budget.

4.3. Scaling with noise severity

Fig. 3 shows error rate as a function of signal-to-noise ratio on the 144-sentence severity sweep, now spanning 20 to -10 dB in eight levels, and Table 4

reports the fitted severity characteristics. The dominant shape is a knee rather than a line: every system holds a nearly flat floor above roughly 10 dB and transitions to near-total failure within a few dB below its knee point, so a single percentage-points-per-dB slope is not a stable summary of these curves. The slope fitted over the original 20 to 0 dB range, for example, moves from -0.26 to -4.69 points per dB for the Qwen-Audio system in Mandarin when the fit is extended to -10 dB, a range dependence that reflects the flat-then-cliff structure rather than any measurement change. We therefore characterise severity response by the signal-to-noise ratio at which the fitted error rate crosses ten percent, with sentence-level bootstrap confidence intervals, together with the maximum fitted slope as a measure of knee sharpness.

The crossing thresholds separate the classes more informatively than slopes. In Mandarin the specialised streaming system crosses ten percent error at 2.5 dB (95% CI 2.3 to 3.3), the telephone specialist at 8.8 dB (8.2 to 9.4), and the two LLM-ASR systems at -0.1 and 0.3 dB, a tolerance advantage of 2.4 to 2.9 dB over the streaming generalist and 7.6 to 8.9 dB over the telephone specialist. In English the ordering is the same but the margins widen: the streaming system crosses at 6.8 dB (5.0 to 7.9) against 2.4 dB (2.1 to 3.4) for Qwen-Audio and 2.0 dB (1.7 to 2.3) for Qwen3-ASR, a 4.4 to 4.8 dB advantage. The LLM-ASR curves also have the sharpest knees, with maximum fitted slopes of 21 to 36 percentage points per dB against 10 to 21 for the specialised systems, so the LLM systems are better read as flat-floor-then-cliff systems than as gradually degrading ones. The mild-noise regime above 10 dB is effectively free of robustness differences in both languages: every system stays within one point of its clean score. Between 10 and 5 dB the specialised systems begin to move first, and by 5 dB the class ordering is established in both languages. The practical implication for procurement testing is that babble probes near each system's knee, in the 0 to 5 dB window here, separate the classes most sharply, whereas a single 5 dB probe ranks systems reliably at the top of this severity range but the LLM-ASR systems are still within noise of each other at that level.

The extension below 0 dB, added in this revision, reveals a regime that

the original five-point ladder could not see. All systems approach total failure, but the LLM-ASR systems fail distinctively: their Chinese error rates exceed 100 percent, reaching 127 percent for Qwen-Audio and 143 percent for Qwen3-ASR at −10 dB, which is only possible when inserted characters outnumber the reference length. Inspection of the transcripts shows long repetition loops and fluent but unrelated text, that is, insertion-heavy hallucination rather than garbled acoustics, a failure mode that a wrong-but-bounded view of error rates cannot represent and one that is particularly dangerous downstream because a hallucinated transcript is grammatical text that a translation stage will happily translate. In the same regime the streaming systems simply return fragments or silence, and their refusal rates rise to 42 to 58 percent (Appendix Appendix F). No system transcribes anything reliably below 0 dB, and the appropriate engineering response in that regime is capture improvement rather than model selection.

The severity sweep also anchors the benchmark in classical signal quality. Short-time objective intelligibility (STOI) computed against the clean signals falls monotonically with noise level, from 0.97 at 20 dB through 0.67 at 0 dB to 0.40 at −10 dB, and within the babble family the condition-level correlation between STOI and recognition error is strong. Across degradation families, however, STOI and error dissociate sharply: reverberation at a measured STOI of 0.63 leaves every generalist system below 3.2 percent error, while 0 dB babble at a comparable STOI of 0.67 produces 25 to 59 percent error, and tempo distortion collapses STOI to 0.16 through temporal misalignment while leaving every system at its clean error rate. Intelligibility-oriented signal metrics therefore do not predict black-box recognition error across degradation families, and service-level benchmarks of this kind measure something the signal-quality literature cannot substitute for.

4.4. Cascade versus end-to-end translation under degradation

Fig. 4 compares the four speech translation paths across the main grid, and Table 5 reports the corresponding scores with both references, sample sizes,

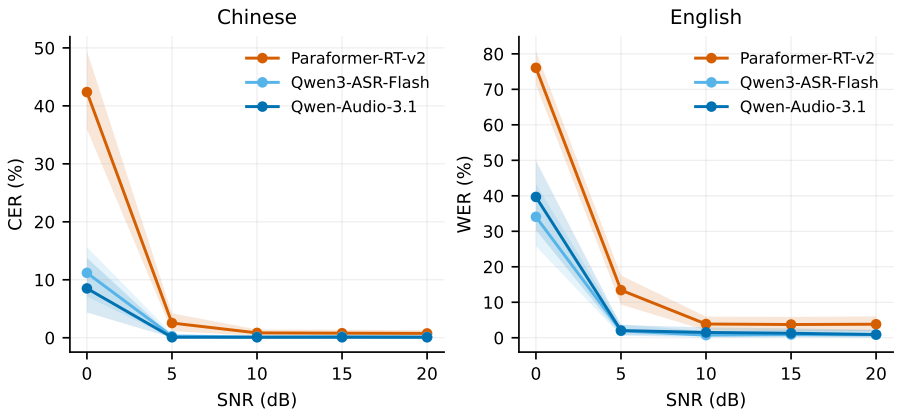


Figure 3: Recognition error rate against babble noise signal-to-noise ratio for four systems and both languages, with bootstrap 95 percent confidence bands, on the 144-sentence severity sweep (72 sentences per language per point). Dashed horizontal line marks the ten percent error level used for the knee thresholds of Table 4.

and the refusal-inclusive chrF. On clean audio the three cascades lose almost nothing relative to their translation oracle, the text translation engine applied to perfect transcripts, which scores 55.5 BLEU against the primary reference. The cascade built on Qwen-Audio 3.1 starts at 53.9 BLEU (n=144), the cascade built on the streaming specialist at 51.5, the cascade built on Qwen3-ASR at 53.2, and the end-to-end omni path scores highest at 68.1 (n=84) with the streaming LiveTranslate service at 59.8 (n=149). Under degradation the ordering and the magnitudes change dramatically. At 0 dB the Qwen-Audio cascade drops 9.8 BLEU points, the streaming-specialist cascade collapses by 35.7 points to 15.8 BLEU, the Qwen3-ASR cascade drops 10.2 to 43.0, the omni end-to-end path loses 22.5 points, and the streaming translation service loses 9.3 points from a lower starting level. The refusal-inclusive chrF tells the same story with the empty outputs counted: the Qwen-Audio cascade falls from 65.2 to 53.7 chrF at 0 dB once its eight refused sweep sentences are scored as empty translations, a drop that corpus BLEU would understate because its brevity penalty saturates when hypotheses are longer than references, which is why both metrics are reported.

Table 4: Noise severity characteristics on the eight-level sweep: fitted signal-to-noise ratio at which error crosses ten percent, with sentence-level bootstrap 95 percent confidence intervals, and maximum fitted logistic slope as a knee-sharpness measure. Sub-range OLS slopes are listed for continuity with common practice and depend strongly on the fitted range.

	SNR at 10% error (dB)		Max slope (pp/dB)		OLS slope, 20–0 dB	
System	zh	en	zh	en	zh	en
Paraformer-RT-v2	2.5 (2.3,3.3)	6.8 (5.0,7.9)	20.8	16.1	−1.44	−2.97
Paraformer-RT-8K	8.8 (8.2,9.4)	n/a	10.4	n/a	−3.31	n/a
Qwen-Audio-3.1	−0.1 (−0.2, 0.6)	2.4 (2.1,3.4)	35.5	21.3	−0.26	−2.79
Qwen3-ASR-Flash	0.3 (−0.1, 1.0)	2.0 (1.7,2.3)	25.2	25.0	−0.35	−2.64

Three conclusions follow. The first concerns the frontend hypothesis. The translation quality of a cascade under acoustic degradation is governed largely by the robustness of its recognition stage: swapping the frontend from the streaming specialist to the LLM recogniser at 0 dB is worth 28 BLEU points on the same backend, and the third cascade, built on Qwen3-ASR, tracks the Qwen-Audio cascade within a point across all twelve shared conditions, which shows that the pattern generalises across frontends of the same class. What the present design cannot separate is the contribution of the translation engine, since every cascade shares one backend; the frontend comparison is within-backend and valid, but a claim that the frontend dominates the choice of engine would require varying the engine too. The second qualifies the common expectation that end-to-end systems avoid cascade error propagation and are therefore more robust. At 0 dB the omni end-to-end path loses a third of its BLEU score, more than the well-frontended cascades lose at 18 and 19 percent, although its absolute score at 45.5 still sits one to two points above the strongest cascade because it started from a much higher clean level. The streaming LiveTranslate service is the most degradation-resilient of the paths, losing 16 percent at 0 dB, which is consistent with the behaviour of its internal frontend recogniser documented in Section 4.7. End-to-end designs do not inherit robustness automatically; they inherit the robustness of whatever acoustic frontend their provider trained

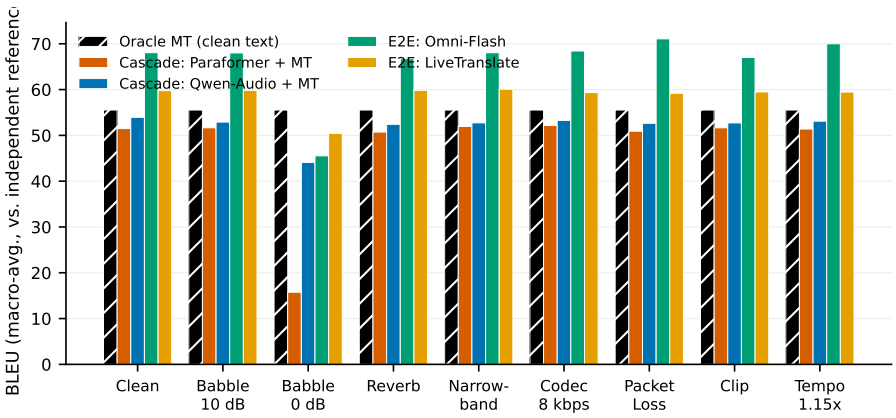


Figure 4: Translation quality of two cascades and two end-to-end speech translation paths across the main degradation grid, scored against the independent LLM reference. The hatched bar is the text-translation oracle on clean transcripts.

internally, which in this measurement is better than the specialised streaming ASR but not automatically better than the best cascade.

The condition-wise structure of Fig. 4 adds three nuances to this picture. First, apart from the 0 dB point the cascade built on the LLM recogniser is essentially flat across the entire grid, moving within one BLEU point of its clean score under every other condition, which mirrors the recognition flatness of Table 3 and confirms that the non-noise degradations are, at deployed severities, engineering noise rather than failure modes for a robust frontend. Second, the streaming translation service shows the same flatness with a single exception: its narrowband score of 60.1 sits marginally above its clean score, and its packet-loss and tempo scores are likewise within half a point of clean, so its 0 dB loss is the only substantive degradation it suffers. Third, the omni end-to-end path exhibits a small but consistent anomaly: its packet-loss and tempo scores, at 71.1 and 70.0, exceed its clean score of 68.1. Inspection of the outputs suggests that under these mild distortions the model produces slightly more concise translations that happen to align better with the reference style, rather than genuinely better translations; the effect is within the reference-sensitivity margin documented in Section 4.5 and is a further argument for treating small

Table 5: Translation quality by condition: macro-averaged BLEU against the primary reference (Qwen3.8-Max translations of clean transcripts) and the secondary reference (oracle cascade output), with the clean-to-0 dB relative loss, refusal-inclusive chrF, and per-cell n. Cascade scores at 0 dB are conditional on a non-empty frontend output; the refusal-inclusive chrF counts refused sentences as empty translations.

	Clean	Babble 0 dB			$\text{chrF}_{0\text{dB}}$	$\text{chrF}_{\text{clean}}$
System	BLEU	BLEU	vs oracle	rel. loss	(incl. empty)	(incl. empty)
Oracle MT (clean text)	55.5	–	–	–	–	66.7
Cascade: Paraformer + MT	51.5	15.8	17.4	69%	26.9	63.4
Cascade: Qwen-Audio + MT	53.9	44.1	68.3	18%	53.7	65.0
Cascade: Qwen3-ASR + MT	53.2	43.0	66.7	19%	53.4	65.4
E2E: Omni-Flash	68.1	45.5	34.4	33%	51.6	73.0
E2E: LiveTranslate	59.8	50.4	44.9	16%	56.2	68.5

n: cascades 144 at clean; at 0 dB 136 (Qwen-Audio), 135 (Paraformer), 144 (Qwen3-ASR) non-empty; omni 84; LiveTranslate 144 clean, 132 at 0 dB (websocket drops).

absolute differences between heterogeneous systems with caution.

Decomposing the 0 dB collapse by direction sharpens the frontend story. The specialist cascade loses 25.9 BLEU points in the Chinese-to-English direction but 45.6 points in the English-to-Chinese direction, falling to 8.7 BLEU, because its English recognition, at 75.5 percent error, destroys the source representation before the translation stage operates. The LLM-frontend cascade degrades far less but still asymmetrically, losing 4.7 points in the Chinese-to-English direction and 15.0 in the other, a 3.2 to one ratio, and the omni path shows the same skew at 19.4 against 25.7. The streaming translation service loses almost nothing in the Chinese-to-English direction at 0 dB, at 56.0 against 57.3 clean, but 18.3 points in the other direction, which parallels the language asymmetry of its internal frontend. No path degrades symmetrically, so direction-symmetric deployment claims should be checked per direction rather than pooled.

4.5. Reference sensitivity of translation scoring

Table 5 also reports scores against the secondary reference, the oracle cascade output. The two references invert the apparent ordering of the systems on clean audio. Against the primary reference, produced by the provider's strongest text LLM, the omni end-to-end path leads at 68.1 BLEU, exceeding even the text-translation oracle at 55.5. Against the oracle reference the same end-to-end path scores 48.5 while the Qwen-Audio cascade scores 89.6, since the cascade shares its translation engine with the reference generator. The inversion repeats at 0 dB: the omni path leads the Qwen-Audio cascade by 1.4 points against the primary reference but trails it by 33.9 points against the oracle reference. The inversion quantifies a hazard for any evaluation that scores machine translation against machine-generated references, and it is sharpened here by the fact that all systems and both reference generators come from one vendor, so no reference in this study is family-disjoint from every system under test; the primary reference at least comes from a different product line than any translation path. Two safeguards make the present conclusions robust to this effect. Degradation losses are computed within a system against a fixed reference, so the percentage losses reported above do not depend on the choice, and the ordering of relative losses at 0 dB, at 69, 33, 19, 18, and 16 percent, is preserved under both references. We recommend that future black-box translation evaluations score against at least two references generated by different model families from each other and from every system under test, state the family relationships explicitly, and treat cross-family absolute comparisons as provisional until human-reference anchored.

4.6. Error pass-through from recognition to translation

Fig. 5 relates the BLEU loss of the three cascades to their mean recognition error rate across all operating conditions, now fifteen per cascade for the two original frontends and twelve for the Qwen3-ASR cascade, spanning recognition error rates from 0.3 to over 100 percent. The error metric on the horizontal axis is the sweep-subset mean recognition error with refusals counted as complete

failures, matching the protocol of Table 3, and the vertical axis is the BLEU loss from each cascade's clean score against the primary reference. The relationship is strongly linear across this range: the Pearson correlation is 1.00 for the Qwen-Audio cascade, 0.995 for the specialist cascade, and 0.994 for the Qwen3-ASR cascade. The fitted pass-through regression with explicit intercept gives slopes of 0.44 BLEU points per percentage point of recognition error for the Qwen-Audio cascade (sentence-level bootstrap 95% CI 0.40 to 0.47), 0.54 for the specialist cascade (0.51 to 0.58), and 0.47 for the Qwen3-ASR cascade (0.36 to 0.65), with intercepts statistically indistinguishable from zero in all three cases. Leave-one-condition-out refits move the first two slopes by less than 0.03, and excluding the high-error conditions above 40 percent recognition error, which carry the most leverage, moves the slope from 0.44 to 0.37 for the Qwen-Audio cascade and from 0.54 to 0.68 for the specialist cascade, so the pass-through rate is approximately but not exactly constant across the error range. The steeper slope for the specialised frontend reflects the composition of its errors under severe noise, which include whole-utterance failures where the translation engine receives little or no usable signal, whereas the LLM recogniser's residual errors under moderate degradation are concentrated in individual words that the translation engine can often route around. A service integrator can therefore use the rule in its calibrated form: every percentage point of recognition error costs roughly half a BLEU point downstream, with the caveat that the cost drifts toward two thirds of a point for error profiles dominated by whole-utterance failures and toward two fifths for error profiles dominated by isolated word errors, and that the estimate is calibrated on one translation engine and should be re-estimated if the backend changes.

4.7. The streaming translation frontend as a recogniser

The streaming translation service exposes its internal recognition output, which allows its frontend, a specialised streaming recogniser, to be scored as a sixth system on the same 144-sentence sweep subset as the directly purchasable systems, which is the comparison reported here. Its clean-condition error rate of

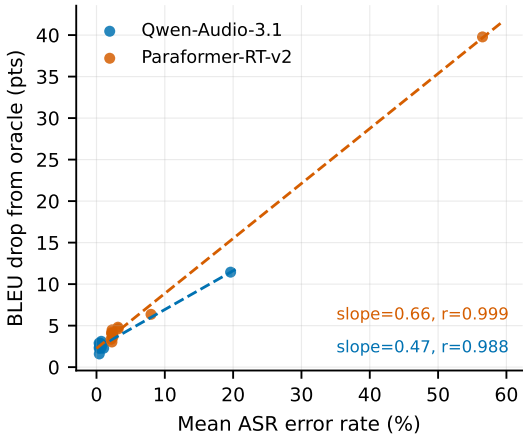


Figure 5: Translation quality loss against mean recognition error rate across fifteen operating conditions for three cascades, with fitted linear pass-through (with intercept) and 95% confidence bands. Recognition error is the sweep-subset mean with refusals counted as complete failures; sub-zero-dB conditions extend the range beyond 100 percent through insertion errors.

1.0 percent sits between the LLM-ASR systems at 0.4 and the general-purpose streaming specialist at 2.2 percent, a difference that is not individually significant, and under noise it tracks the LLM-ASR class rather than its own product class: at 0 dB it errs at 25.4 percent against 24.1 and 22.6 percent for the two LLM-ASR systems, with overlapping confidence intervals and no significant paired difference (Wilcoxon $p = 0.19$), while the general-purpose streaming specialist errs at 59.2 percent on the same sentences. At 2.5 dB it errs at 4.0 percent against 4.5 and 2.6 for the LLM systems. Two coverage caveats apply: the frontend is only observable where the translation session completed, so its sweep coverage is partial in some conditions (52 to 134 of 144), and its zero-error 5 dB figure in the appendix rests on 52 completed sessions and should not be read as outperforming its own clean score. The observation remains noteworthy for practitioners: a recogniser competitive with the best purchasable ASR of this provider under severe noise is not available as a recognition service but is embedded inside a translation product, which suggests that providers optimise internal streaming frontends for exactly the acoustic conditions where

end-to-end translation must survive.

4.8. Sentence-level variability and latency

Fig. 6 shows the sentence-level error distributions for the strongest LLM recogniser and the specialised streaming recogniser on five representative conditions. The distributions are strongly right-skewed: under moderate degradation most sentences remain nearly correct while a tail of sentences fails completely, which is why the confidence intervals in Table 3 are wide relative to their means. Under 0 dB babble the median sentence is still recognised reasonably well by the LLM systems while the upper quartile fails outright, a heterogeneity that aggregate error rates conceal and that matters for applications with per-utterance quality requirements. The skew also explains a statistical property visible throughout Table 3: the confidence intervals of the LLM systems under moderate degradation are wide relative to their means because a small number of complete failures, including the refusal cases, coexist with a large mass of near-perfect transcripts. For a system operator this shape matters more than the mean, since the tail sentences are the ones that generate complaints and downstream errors, and the fraction of sentences above a given error threshold is a more actionable deployment metric than the average. On the 0 dB condition, the fraction of English sentences with error rates above 50 percent is 83 percent for the specialised streaming recogniser against 36 percent for Qwen3-ASR, while in Chinese the same threshold is exceeded by 33 percent of sentences for the specialised system against 3 percent for Qwen-Audio, so the family gap in tail behaviour is considerably larger than the gap in means.

Table 6 reports whole-utterance turnaround latencies. The text translation stage, benchmarked separately with 120 sequential timed calls, completes in 0.35 seconds per sentence on average (median 0.26, 95th percentile 0.46), so the fastest cascade that this provider can assemble, Qwen3-ASR feeding the translation engine, completes in about 0.7 seconds per sentence on average, the Qwen-Audio cascade in about 1.2 seconds, and the omni end-to-end path averages 2.8 seconds with a 95th percentile above 7 seconds, reflecting the substan-

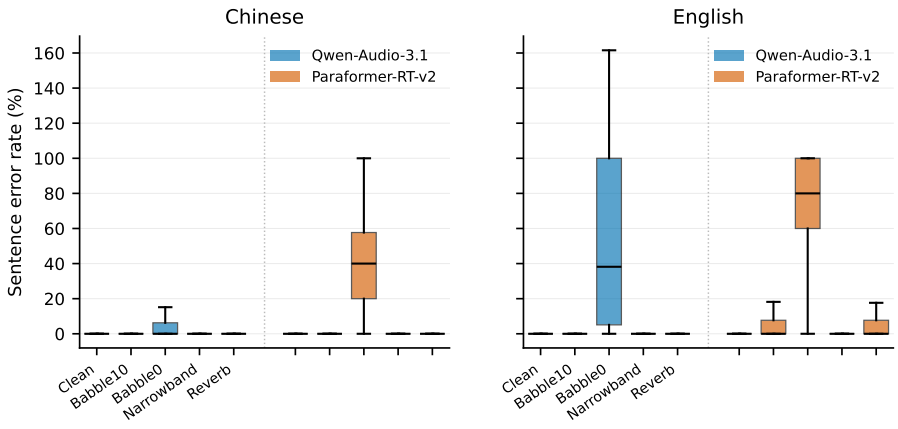


Figure 6: Sentence-level error distributions by condition for the strongest LLM recogniser and the specialised streaming recogniser.

tially longer generation path of an omni-modal foundation model that produces its translation in a single response. The streaming translation service completes in about half a second per utterance. These figures are whole-utterance turnaround, not first-partial latency: the streaming interfaces would show substantially lower time-to-first-text than the whole-utterance numbers suggest, so latency comparisons across interface classes should be treated as indicative. Latency therefore reinforces rather than reverses the accuracy picture: the paths that degrade least under noise, the well-frontended cascades and the streaming translation service, are also among the fastest, while the slowest path, the omni end-to-end model, is the most fragile under severe noise in this measurement.

The repetition runs quantify how much of the measured variation is run-to-run noise rather than signal. Re-running the two LLM-ASR systems twice on the 0 dB babble and tempo conditions yields condition-level error rates that vary by at most 0.9 percentage points across three runs (0 dB: 24.1, 24.9, and 24.4 for Qwen-Audio; 22.6, 23.4, and 23.2 for Qwen3-ASR), while individual transcripts agree in 79 to 99 percent of sentences between runs. Condition-level conclusions in this paper are therefore stable against decoding nondeterminism at default settings, but sentence-level deployments should expect one in five

Table 6: Whole-utterance API turnaround latency in seconds for the recognition systems, the text translation stage (120 sequential timed calls), and the end-to-end paths. Streaming interfaces are measured at maximum feed rate; see text for the first-partial caveat.

System	Mean	P50	P95
Qwen3-ASR-Flash	0.37	0.35	0.46
Paraformer-RT-8K-v2	0.45	0.43	0.75
LiveTranslate (E2E, streaming)	0.55	0.52	0.80
Qwen-Audio-3.1-Flash	0.80	0.72	1.93
Paraformer-RT-v2	0.87	0.85	1.65
Text translation (qwen-mt-uni, n=120)	0.35	0.26	0.46
Qwen3.8-Omni-Flash (E2E)	2.77	2.20	7.19

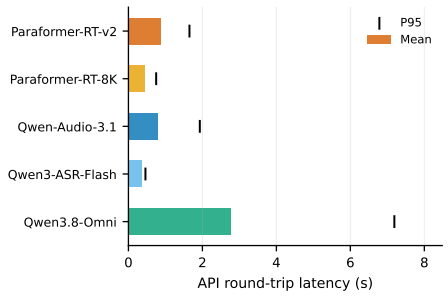


Figure 7: API round-trip latency by system, mean and 95th percentile.

transcripts to differ between identical calls at the worst condition. The omni path's small packet-loss and tempo anomalies of Section 4.4, two to three BLEU points, exceed any error-rate variation we can measure on the systems we could re-run, but the omni system's quota was exhausted before it could be re-tested, so its anomaly remains unexplained rather than confirmed as genuine.

5. Discussion

5.1. Guidance for system integrators

The measurements translate directly into selection guidance for practitioners who compose commercial speech services. Above a signal-to-noise ratio of

about 10 dB, every recognition system measured here operates within one point of its clean error rate on clearly articulated speech, and the choice among them can be driven by latency, cost, and interface ergonomics rather than accuracy. The latency data in Table 6 then favour the audio-input LLM systems and the specialised streaming recognisers, which all answer in well under a second, over the omni-modal path. Between 0 and 5 dB, the product classes separate, and the knee thresholds of Table 4 express the difference most usefully: the LLM-ASR systems cross ten percent error 2.4 to 2.9 dB later in Mandarin and 4.4 to 4.8 dB later in English than the specialised streaming system, so in intermittently noisy environments such as public spaces, vehicles, or shared offices, the foundation-model recognisers hold the better end of the accuracy-latency trade-off. Below 0 dB, every system fails, the LLM systems by hallucination and the streaming systems by silence, and the appropriate response is architectural: capture improvement, multi-microphone processing, or explicit rejection logic. For telephone-band audio, the surprising result is that no specialist premium exists: the telephone specialist matches neither the generalists' accuracy in its own band nor their robustness elsewhere, and is uniquely fragile under reverberation, so its selection can be justified only by its 8 kHz interface requirements and its speed, not by accuracy.

For translation pipelines, the frontend choice dominates the measurements reported here, and the latency data combine with it into a compact decision rule. A cascade assembled from the fastest recognition tier and the text translation engine completes in about 0.7 seconds per sentence on average, roughly four times as fast as the omni end-to-end path, while the streaming translation service completes in about half a second and is therefore the natural choice when latency dominates; when noise robustness dominates, the same streaming service and the LLM-frontend cascades lead the field, so the two objectives point to the same shortlist. Swapping the recognition stage of a cascade from the specialised streaming system to the LLM recogniser is worth 28 BLEU points at 0 dB on an identical translation backend, a far larger effect than any measurement noise in this study, though the present design holds the translation engine fixed

and therefore cannot rank frontend changes against engine changes. The pass-through regression of Section 4.6, at roughly half a BLEU point per percentage point of recognition error with a statistically zero intercept, allows an integrator to rank candidate frontends with recognition-only measurements, which are cheaper to run and easier to interpret than full-pipeline evaluations, and to forecast translation quality degradation from a measured or contracted recognition error rate, with the caveat that the coefficient is calibrated on one translation engine. The end-to-end versus cascade decision should rest on measured degradation behaviour rather than architectural intuition: the omni-modal end-to-end path evaluated here is simultaneously the slowest and, under severe noise, the most fragile of the four translation designs, while a dedicated streaming translation service achieves both the smallest relative loss at 0 dB and sub-second latency.

5.2. Why foundation-model recognisers degrade more slowly

The robustness advantage of the LLM-ASR systems echoes the finding of Radford et al. [9] that recognition models trained on very large and weakly curated corpora acquire substantial noise tolerance without explicit multi-condition training. Two of our observations sharpen this picture for API consumers. First, the advantage is strongly condition-dependent: it is large under energetic babble noise and negligible under codec, packet-loss, clipping, and tempo distortions, which barely affect any system. This pattern is consistent with robustness arising from exposure to diverse recording conditions during pre-training rather than from any explicit enhancement front-end. Second, the advantage is language-asymmetric, with Chinese recognition remaining usable at 0 dB while English recognition of the same systems fails at 37 to 50 percent, and, as Section 4.3 notes, part of the contrast may reflect the character-level versus word-level scoring scales rather than raw robustness alone. Providers disclose neither training mixture nor per-language capacity, so the asymmetry can only be documented, not explained, at the black-box level; the documentation itself is actionable, since it implies that multilingual deployments should be derated

per language rather than per provider. It also implies a concrete procurement test: a noise probe around each system's knee, a few hundred sentences of the target language in the 0 to 5 dB window, places a candidate service on the robustness scale documented here, and repeating the probe per deployment language guards against extrapolating a strong Mandarin result to a weak English result or the reverse. The direction decomposition of Section 4.4 extends the same caution to translation deployments, where the two directions of a bilingual service degraded by asymmetric amounts, from 1.3 to 1 ratio to 3.2 to 1, under identical noise.

5.3. Cascade versus end-to-end revisited

The classical comparison of Bentivogli et al. [12] found end-to-end and cascade systems competitive on clean speech with different error structures. Our measurements extend that comparison to degraded channels in the black-box setting and qualify it in two ways. The end-to-end advantage in absolute quality against an independent reference is real but partly a family artefact, since the omni system that translates directly also shares a generation family with the strongest reference producer, and the advantage inverts under the secondary reference, as Section 4.5 shows. More importantly, robustness is not an architectural property but a property of the acoustic frontend wherever it lives. The end-to-end service with the specialised internal frontend is the most resilient path we measured, while the end-to-end omni path is the most fragile, and the same frontend ordering that explains cascade behaviour also explains end-to-end behaviour. For a consumer, the practical question is not which architecture to prefer but where the provider placed its best acoustic model, and the answer is discoverable only by measurement of the kind this benchmark performs.

5.4. Methodological implications for black-box evaluation

Three methodological observations generalise beyond the specific services tested. The dual-reference protocol of Section 4.5 demonstrates that machine-generated evaluation references can reorder systems in a way that tracks reference-generator affinity rather than translation quality, and that degradation-relative

conclusions are the stable currency of black-box comparison. The synthesis-based corpus design gives black-box studies an exact transcript by construction, which matters because at sub-percent clean error rates, transcription uncertainty in recorded corpora would be comparable to the effects being measured. Finally, the empty-output behaviour documented in Section 4.1, where three of the five systems return empty transcripts on a systematic subset of severely degraded utterances, through two distinct mechanisms, argues that black-box benchmarks should report response completeness alongside error rates, since a missing transcript and a wrong transcript have different downstream failure modes, and a silently empty one differs operationally from a protocol-level refusal.

5.5. A worked example of the pass-through rule

The pass-through regression is most useful when stated as an explicit calculation, including its intercept. For the Qwen-Audio cascade the fitted relation is a loss of $0.44\,e + 0.6$ BLEU points at recognition error e percent. Suppose an integrator operates that cascade in an environment that occasionally degrades to 10 dB, at which the recogniser errs at 0.8 percent on the sweep subset; the predicted translation loss is 0.9 BLEU points, which is below the clean-condition agreement between the two reference systems and therefore negligible, and consistent with the measured 1.0 point loss at that condition. If the same deployment used the specialised streaming recogniser, whose 10 dB error rate is 2.4 percent on the same sentences, the specialist-calibrated regression predicts $0.54 \times 2.4 - 1.0$, about 0.3 BLEU points, again matching the measured figure within a point. In the 0 dB regime the rule predicts a loss of $0.44 \times 24.1 + 0.6$, about 11 points, against a measured 9.8, and for the specialist frontend $0.54 \times 56.5 - 1.0$, about 29 points, against a measured 35.7, so the rule is accurate to within a few points up to the edge of the usable regime and understates the collapse beyond it, which is exactly where the leverage check of Section 4.6 shows the slope rising toward two thirds of a point. The same arithmetic applied to the measured error rates of a candidate system at the

actual operating signal-to-noise distribution of a deployment yields an expected translation quality loss without running the translation stage at all, and its error bars, roughly ± 0.04 on the slope, are small next to the business consequences of the frontend decision it informs.

5.6. Implications for evaluation practice and future work

Beyond the specific findings, the benchmark design carries lessons for future evaluations of commercial speech intelligence. A benchmark intended to track black-box services over time should log every raw response rather than only aggregate scores, because behavioural changes such as the refusal pattern documented here appear as distributional shifts before they appear in means. It should treat reference construction as a first-class experimental variable, since Section 4.5 shows that the reference generator can reorder systems. It should separate degradation gradients, which generalise across severity, from point estimates, which do not, and it should report response completeness alongside accuracy. The present study points to four directions for follow-up work: replication across providers to separate family effects from provider effects, extension to spontaneous and accented speech, sweep-based characterisation of the currently fixed-severity conditions, and human-reference scoring of the translation stage to anchor the machine-reference comparisons. As foundation model services begin to expose streaming speech interfaces with embedded recognition, the measurement approach demonstrated here, treating each service as an opaque function from degraded audio to text and scoring the complete pipeline end to end, is, at present, the most direct methodology available for the comparative robustness questions that practitioners actually face.

6. Limitations

The benchmark uses synthetic speech from four neural voices, each reading each sentence once, so voice-level and take-level variability is not sampled and the corpus is read-style by construction. Synthetic speech lacks the disfluencies,

accents, emotional variation, and channel noise of spontaneous human recordings, and although the exact-transcript property is decisive for measurement at sub-percent error floors, clean error rates measured on synthetic speech should not be read as population estimates for human speech; the degradation deltas, which are the object of this study, are less sensitive to this concern because they are within-system contrasts, but a real-speech replication remains necessary before the clean-condition levels can be generalised. The most consequential design constraint is that all systems, references, and the synthesis come from a single provider, so product class and vendor are fully confounded: the class-level regularities reported here are consistent with published training-scale explanations [9], but cross-provider replication is required before they can be attributed to architecture rather than to one provider's engineering, and the same confound applies to the reference translations, since no reference generator family-disjoint from all systems was available; human references or references from other vendors are needed to anchor absolute translation quality. The babble noise is itself synthesised from disjoint neural voices, which makes it speech-shaped and fully reproducible but smoother than real multi-talker cocktail-party interference, which includes semantic competition and stronger nonstationarity. The reverberation condition uses a synthetic exponentially decaying response whose diffuse tail yields a direct-to-reverberant ratio of -27.6 dB at a moderate clarity ratio, which under-represents the discrete early reflections of real small rooms; real recorded impulse responses and real environmental noise would strengthen the acoustic realism of both categories. Severity levels for the non-noise degradations were fixed at deployed-but-aggressive values rather than swept, so only the noise family is characterised by a curve, and the Opus and narrowband conditions proved to be partially redundant stressors at the chosen bitrate. Translation quality is measured with BLEU and chrF against machine-generated references, with the family-affinity bias that Section 4.5 quantifies; no learned reference-free metric (COMET class) or human judgement was available within the zero-cost constraint, and human-reference scoring is the single addition that would most strengthen the absolute claims. The corpus sentences were authored

to be neutral and predictable, so the language-model prior of the LLM recognisers may partly mask acoustic degradation, and the numerals-domain gap may partly reflect inverse text normalisation rather than acoustics, as the normalisation sensitivity analysis suggests. Finally, commercial APIs are moving targets: the measurements document the October 2026 versions of the listed services, one system could not complete its full condition grid before its free-tier quota was exhausted, and the websocket transport of the streaming translation service dropped roughly ten percent of its sessions, which is itself a practically relevant datum for anyone planning a similar measurement on complimentary quotas.

7. Conclusions

This paper introduced CascadeBench, a black-box benchmark that measures how acoustic channel degradations propagate through commercial speech recognition and translation services, and reported a complete measurement over a controlled bilingual corpus of 480 sentences, eight degradation conditions, and a five-level noise severity sweep, covering five recognition systems from three architectural families, two cascade translation paths, and two end-to-end speech translation paths, all accessed through their public interfaces on free service quotas.

The measurements support five conclusions. First, the product classes of commercial recognisers differ in where they fail under noise, not merely how fast: LLM-based recognisers hold a sub-percent floor and cross ten percent error 2.5 to 2.9 dB later in Mandarin and 4.4 to 4.8 dB later in English than the specialised streaming system, then collapse more sharply, and below 0 dB they fail by insertion-heavy hallucination with character error rates above 100 percent. Second, robustness is condition- and language-specific rather than uniform: narrowbanding, coding, packet loss, clipping, and tempo changes at deployed severities are individually negligible for all systems, 0 dB babble nearly destroys English recognition for every system while leaving the best Mandarin recognition low, and no system, of any class, transcribes reliably below 0 dB.

Third, the quality of a speech translation cascade under degradation is governed by its recognition frontend, and the relationship is linear and predictable, with a pass-through of 0.44 to 0.54 BLEU points per percentage point of recognition error, bootstrap confidence intervals within a few hundredths, a statistically zero intercept, and stability under leave-one-condition-out refits, across fifteen conditions and three frontends. Fourth, end-to-end speech translation does not inherit robustness automatically: the most resilient translation path measured is a dedicated streaming end-to-end service, while the omni-modal end-to-end path is both the slowest and, under severe noise, the most fragile, even though its absolute score remains highest at every condition measured. Fifth, black-box evaluation carries specific methodological obligations that this benchmark operationalises, including dual-reference scoring to control machine-reference affinity, exact-transcript corpora to enable sub-percent measurement, the reporting of response completeness alongside error rates, and the anchoring of service-level findings in measured signal properties, which here exposed a sharp dissociation between intelligibility metrics and recognition error across degradation families.

The benchmark recipe, including the corpus, the seeded degradation pipeline, the degraded audio tree, all model responses, and the analysis code, is released so that the measurement can be re-executed and verified as providers update their services. The released artefacts are versioned alongside the service identifiers and access dates of every system, so that future re-runs can attribute score movements either to provider-side model updates or to changes in the benchmark itself, a distinction that periodic evaluation of moving targets requires. Together with the zero-cost execution protocol described in Section 3, the release is intended to make robustness re-measurement a routine maintenance activity for speech deployments rather than a one-off research exercise. As speech intelligence continues to migrate into provider-operated black-box systems, controlled measurement of this kind, repeated over time and across providers, offers the community a practical substitute for the training-set access that classical robustness research assumed, and offers integrators the quantitative basis that architectural intuition alone cannot supply.

Appendix A. Corpus composition

Table A.7 summarises the corpus structure and lists one sample sentence per stratum. The four domains are constructed to differ in register, vocabulary, and numerical density. The broadcast-news domain uses formal written register with institutional vocabulary, the conversation domain uses colloquial everyday phrasing, the technical domain uses scientific and engineering terminology, and the numerals-and-entities domain concentrates dates, times, quantities, prices, addresses, codes, and named entities, with all numbers written out in words as described in Section 3. Length strata are defined per language as short, medium, and long as given in Section 3, and every domain-by-length cell contains twenty sentences per language, giving 240 sentences per language and 480 in total. The severity sweep subset of 144 sentences is sampled evenly across all cells, six sentences per cell per language.

Table A.7: Corpus structure with one sample sentence per domain.

Domain	Sample sentence
News (zh)	国家统计局今天发布了上月经济运行数据。
News (en)	The central bank raised interest rates by twenty five basis points to curb inflation.
Conversation (zh)	我打算下班之后去健身房锻炼一个小时。
Conversation (en)	I was thinking we could drive to the beach this weekend if the weather holds.
Technical (zh)	混响是声波在房间内多次反射形成的，会显著降低远场语音识别的准确率。
Technical (en)	Reverberation arises when sound waves reflect repeatedly off room surfaces.
Numerals (zh)	从北京到上海的高铁大约需要四个半小时。
Numerals (en)	The high speed train from Beijing to Shanghai takes about four and a half hours.

Appendix B. Qualitative error analysis

Table B.8 collects representative model responses that illustrate the failure patterns behind the aggregate numbers in Section 4.

The refusal behaviour of Section 4.1 is shown in the first row: under 0 dB babble the Qwen-Audio recogniser returns an empty transcript for a substantial English sentence, which the benchmark scores as a complete failure. The second row shows the opposite regime: the same system, on a long English news sentence at the same severity, produces a verbatim transcript, which underlines how bimodal the sentence-level distributions of Fig. 6 are. Row three shows the dominant error mode of the specialised streaming recogniser under severe noise, where most of the sentence is lost and only a short fragment survives. Row four shows a typical near-miss of the LLM recogniser in Chinese, where a single content word, density, is replaced by another word of similar phonetic shape, speed, without any structural damage. Row five shows the collapse of the telephone-band specialist under reverberation, which occasionally produces short disconnected fragments rather than transcripts. Row six shows an error that the recognition stage never sees but the translation consumer experiences: under 0 dB noise the omni end-to-end path substitutes a plausible but wrong entity, rendering the sports centre of the source as a grand theatre, a hallucination that a string-matching metric penalises only lightly but that is catastrophic for factual fidelity. Row seven shows the narrowband immunity of the LLM recogniser, which returns a verbatim transcript of telephone-band audio. Together the examples support the reading of Section 5: the systems differ less in whether they fail than in how, and the failure modes, from verbatim robustness to silent entity substitution to outright refusal, have distinct downstream consequences.

Appendix C. Cross-system text normalisation

Comparing transcripts across commercial recognisers requires a normalisation protocol that neutralises provider-specific inverse text normalisation. The protocol applies the following rules identically to references and hypotheses, in

order. Unicode compatibility folding and case folding are applied first, and punctuation is removed. Numeric expressions are then canonicalised to digit form on both sides. For Chinese, the rules handle cardinal numbers with the standard multipliers, digit-by-digit readings such as room and code numbers where each digit is read separately, decimal fractions written with the point particle, time-of-day expressions in which the hour and minute parts are converted separately, percentage markers, which are deleted on both sides, and scale multipliers, where a digit followed by wan, yi, or their combinations is expanded to the full multiplied-out form, so that twenty-six million written as 2600 万 in one system's output and as 26000000 in another's both canonicalise identically. For English, the rules handle cardinal numbers including compound hyphenated forms and the and connector, time and digit-by-digit readings such as twelve zero eight and nine forty, which canonicalise to 1208 and 940, decimal fractions expressed with point, scale multipliers from thousand to trillion, and ordinal expressions of both word and digit form, which are masked to a common placeholder since their spoken-to-written realisations differ across systems in ways that carry no acoustic information. The rule set was developed and frozen by inspecting all high-error sentences of the clean condition until no residual mismatches attributable to normalisation remained, a procedure that could in principle favour whichever system's conventions dominate the clean outputs, so a sensitivity analysis was run: re-scoring the main grid with case and punctuation folding only, dropping numeral canonicalisation entirely, raises clean-condition error rates from 0.5, 2.3, and 0.3 percent to 6.1, 6.2, and 0.8 percent for Qwen-Audio, Paraformer, and Qwen3-ASR respectively, while the 0 dB scores move by less than 1.5 points and the system ordering is unchanged in every condition. The normalisation therefore compresses but does not create the reported system differences, and the residual protocol noise on clean speech is bounded above by the strongest system's per-language clean rate of 0.14 percent in Mandarin (0.29 percent pooled over both languages), with the caveat that the two systems most affected by numeral conventions differ most under the raw protocol. The full normalisation code is released with the benchmark.

Appendix D. Complete pairwise significance results

Table D.9 reports the pairwise Wilcoxon signed-rank comparisons between recognition systems that remain significant at the Holm-corrected level within the family of all pairwise comparisons of the same condition and language, so the table lists only significant results by construction. Two protocol notes are needed to read it correctly. First, comparisons among the three general-purpose recognisers are computed on the full 480-sentence main grid, 240 sentences per language, whereas every comparison involving the omni system is computed on the 144-sentence sweep subset, 72 per language, because that system never ran the full grid; the two blocks of the table are same-subset within themselves, and the omni rows state their subset explicitly. Second, the fraction of zero paired differences is substantial in mild conditions, 72 to 83 percent under reverberation and narrowband, so significance there is carried by the minority of sentences where the systems differ. The consistency of the ordering is itself informative: within every condition and language, the two LLM-ASR systems occupy the leading positions and the specialised streaming recogniser the trailing position, with the exception that Qwen3-ASR leads Qwen-Audio 3.1 in English under reverberation, clipping, narrowband, and 0 dB noise, where the refusal behaviour of the latter penalises it, while the omni system is statistically inseparable from the Qwen systems in English under noise and reverberation on its subset.

Appendix E. Numeric results of the severity sweep

Table E.10 reports the complete error rates of the eight-level severity sweep with confidence intervals, and Table E.11 reports the internal frontend recogniser of the streaming translation service on the same subset. Three details of the sweep data deserve note. The confidence intervals at 0 dB and below are wide for every system, spanning ten percentage points or more, which reflects the bimodal sentence-level behaviour documented in Section 4.8. The crossover between the classes occurs between 5 and 0 dB for both languages, and at 5 dB the specialised system already shows its first significant departure from

its clean score while the LLM-ASR systems remain within 0.3 points of theirs in Mandarin. The internal frontend of the streaming translation service has partial coverage in some sweep conditions, 52 to 134 of 144 sessions, because its websocket transport dropped sessions that did not recover within the retry budget; its zero score at 5 dB rests on 52 completed sessions and is a small-sample fluctuation rather than evidence of an operating point better than clean. The sub-zero conditions, 2.5, −5, and −10 dB, were added in the revision round with identically seeded babble realisations, and their per-point sample sizes equal those of the original ladder, 72 sentences per language per level.

Appendix F. Response completeness: refusals and empty transcripts

Table F.12 reports the fraction of utterances for which each system's final response contains no transcript, split by the two mechanisms: protocol-level refusals, where the API returns an explicit no-words error, and silent empties, where a successful response carries an empty string. The distinction matters operationally: an explicit error can be caught by the caller and retried or escalated, whereas a silent empty return is indistinguishable from a genuinely silent audio segment. In the main grid only the 0 dB condition produces empty outputs; in the sub-zero extension they become a dominant failure mode for the streaming systems, with 42 to 58 percent of utterances returning nothing at −5 dB, while the LLM systems increasingly return hallucinated text instead. No failed API call, of any kind, ever recovered to a different outcome on retry within the six-attempt budget, and the 24 protocol refusals of the Qwen-Audio system at 0 dB were stable across repetition runs, so the behaviour is deterministic in the input rather than an artefact of transient service conditions.

Data and code availability

The complete benchmark is publicly released at https://github.com/43aquarius/CascadeBench-Paper-AppliedAcoustics: the sentence corpus, the seeded perturbation implementations, the full degraded audio tree, all per-sentence model

transcripts and translations (33,000 logged API responses), the analysis scripts, and the LaTeX source of this manuscript. Every number and figure in the paper is regenerated from the released logs by the released analysis scripts. An anonymised snapshot for reviewers is provided with the submission.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large language model assistant in the development of the analysis scripts and in drafting revisions of the manuscript text, in order to accelerate the benchmark construction and to improve the readability of the presentation. The experimental measurements, the analysis decisions, and the verification of all reported numbers against the released logs were performed and checked by the authors, who take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table B.8: Representative system responses under degradation.

Condition	System	Reference and response
0 dB, en	Qwen-Audio-3.1	Reference: Police arrested two suspects on Tuesday. Response: empty transcript.
0 dB, en	Qwen-Audio-3.1	Reference: The transport minister told reporters on Friday that the government would invest heavily in electric vehicle charging infrastructure. Response: verbatim transcript.
0 dB, en	Paraformer-RT-v2	Reference: The battery lasts twenty hours. Response: About.
0 dB, zh	Qwen-Audio-3.1	Reference: 这款处理器的晶体管密度提升了近一倍。Response: 这款处理器的晶体管速度提升了近一倍。
Reverb, zh	Paraformer-RT-8K	Reference: 记者从发布会上了解到，本次合作项目将覆盖全国二十多个主要城市。Response: 七月同。非常了解到。
0 dB, E2E	Omni-Flash	Source: 新建成的体育中心可容纳六万名观众。Translation: The newly built Grand Theatre can accommodate 60,000 audience members.
Narrow, en	Qwen-Audio-3.1	Reference: This coffee shop is cozy. Response: verbatim transcript.
−10 dB, en	Qwen-Audio-3.1	Reference: The battery lasts twenty hours. Response: fluent unrelated Chinese text of several sentences (hallucination).

Table D.9: Significant Holm-corrected pairwise Wilcoxon comparisons by condition and language. Cells give the mean error rates in percent of the two systems compared. Upper block: full main grid, n=240 per language. Lower block: 144-sentence sweep subset, n=72 per language (omni system).

Condition, language	Comparison	Means (%)	p
0 dB, zh	Qwen-Audio vs Paraformer-RT	6.3 vs 41.6	< 0.001
0 dB, zh	Qwen3-ASR vs Paraformer-RT	13.1 vs 41.6	< 0.001
0 dB, zh	Qwen-Audio vs Qwen3-ASR	6.3 vs 13.1	< 0.001
0 dB, en	Qwen-Audio vs Paraformer-RT	50.2 vs 75.5	< 0.001
0 dB, en	Qwen3-ASR vs Paraformer-RT	37.1 vs 75.5	< 0.001
0 dB, en	Qwen-Audio vs Qwen3-ASR	50.2 vs 37.1	< 0.001
Reverb, zh	Qwen-Audio vs Paraformer-RT	0.2 vs 1.3	< 0.001
Reverb, zh	Qwen3-ASR vs Paraformer-RT	0.1 vs 1.3	< 0.001
Reverb, en	Qwen3-ASR vs Paraformer-RT	0.5 vs 5.1	< 0.001
Reverb, en	Qwen-Audio vs Qwen3-ASR	1.2 vs 0.5	0.011
Narrow, zh	Qwen-Audio vs Paraformer-RT	0.1 vs 0.7	< 0.001
Narrow, zh	Qwen3-ASR vs Paraformer-RT	0.1 vs 0.7	< 0.001
Narrow, en	Qwen3-ASR vs Paraformer-RT	0.5 vs 3.6	< 0.001
Narrow, en	Qwen-Audio vs Paraformer-RT	1.0 vs 3.6	< 0.001
Narrow, en	Qwen-Audio vs Qwen3-ASR	1.0 vs 0.5	0.015
Clip, en	Qwen3-ASR vs Paraformer-RT	0.4 vs 4.4	< 0.001
Clip, en	Qwen-Audio vs Paraformer-RT	1.6 vs 4.4	< 0.001
Clip, en	Qwen-Audio vs Qwen3-ASR	1.6 vs 0.4	< 0.001
Sweep-subset comparisons involving the omni system (n=72 per language):			
0 dB, zh	Qwen-Audio vs Omni	8.5 vs 20.2	< 0.001
0 dB, zh	Qwen3-ASR vs Omni	11.2 vs 20.2	< 0.001
0 dB, zh	Paraformer-RT vs Omni	42.4 vs 20.2	< 0.001
0 dB, en	Qwen3-ASR vs Omni	34.1 vs 43.3	0.018
0 dB, en	Paraformer-RT vs Omni	76.1 vs 43.3	< 0.001
Reverb, zh	Paraformer-RT vs Omni	1.3 vs 0.1	0.003
Reverb, en	Paraformer-RT vs Omni	5.0 vs 2.2	0.002

Table E.10: Recognition error rates in percent on the severity sweep subset, 144 sentences per level, with bootstrap 95 percent confidence intervals. Values above 100 percent arise from insertion-dominated outputs.

System	Lang	20	15	10	5	2.5	0	−5	−10
Paraformer-RT-v2	zh	0.8	0.8	0.8	2.5	9.8	42.4	101.2	101.9
		(0.3,1.3)	(0.3,1.4)	(0.3,1.4)	(1.1,4.2)	(6.3,13.9)	(36.0,49.5)	(98.6,104)	(99.5,104)
	en	3.8	3.7	3.9	13.4	38.1	76.1	99.5	99.8
		(1.9,6.0)	(1.9,5.9)	(2.1,6.0)	(9.3,17.6)	(31.4,45.3)	(71.2,80.8)	(98.4,101)	(99.1,101)
Qwen-Audio-3.1	zh	0.1	0.1	0.1	0.1	0.6	8.5	127.1	138.7
		(0.0,0.2)	(0.0,0.2)	(0.0,0.2)	(0.0,0.2)	(0.1,1.3)	(4.3,13.8)	(120,136)	(132,148)
	en	0.9	1.3	1.5	2.0	8.5	39.7	102.6	105.8
		(0.1,2.3)	(0.3,2.6)	(0.4,2.9)	(0.8,3.6)	(4.6,13.4)	(30.3,49.9)	(99.3,107)	(102,110)
Qwen3-ASR-Flash	zh	0.1	0.2	0.1	0.3	1.0	11.2	104.8	142.9
		(0.0,0.4)	(0.0,0.5)	(0.0,0.4)	(0.0,0.7)	(0.3,2.1)	(7.1,15.7)	(98,112)	(135,151)
	en	0.9	0.9	0.7	1.9	4.3	34.1	101.6	104.7
		(0.0,2.2)	(0.1,2.2)	(0.0,1.9)	(0.6,3.8)	(2.0,7.5)	(25.8,43.4)	(98.5,105)	(102,108)

Table E.11: Error rate in percent of the streaming translation service internal frontend recogniser, with session counts; coverage is partial where the websocket transport dropped sessions.

Condition	Clean	20 dB	15 dB	10 dB	5 dB	2.5 dB	0 dB	−5 dB
Error (%)	1.0	0.8	0.2	0.9	0.0	4.0	25.4	100.8
95% CI	(0.3,1.7)	(0.3,1.6)	(0.0,0.6)	(0.3,1.6)	(0.0,0.0)	(1.9,6.7)	(19.8,31.4)	(98.4,104.1)
n	144	134	64	144	52	144	144	107
Condition	Reverb	Narrow	Codec	Loss	Clip	Tempo	−10 dB	
Error (%)	0.8	0.8	0.7	0.8	0.8	1.0	—	
95% CI	(0.3,1.5)	(0.3,1.6)	(0.2,1.4)	(0.3,1.5)	(0.3,1.6)	(0.4,1.8)	—	
n	144	144	144	144	142	142	0	

Table F.12: Empty final transcripts in percent of evaluated utterances, by mechanism (R: protocol refusal; S: silent empty). Main grid conditions report the full 480-sentence grid (240 per language); sub-zero conditions report the 144-sentence sweep subset (72 per language).

System	Lang	2.5 dB	0 dB	−5 dB	−10 dB	Reverb	Mechanism
Paraformer-RT-v2	zh	4.2 (S)	2.5 (S)	41.7 (S)	43.1 (S)	0	silent
	en	4.2 (S)	15.0 (S)	58.3 (S)	52.8 (S)	0	
Paraformer-RT-8K	zh	1.4 (S)	0.8 (S)	8.3 (S)	15.3 (S)	0.8 (S)	silent
Qwen-Audio-3.1	zh	0	0.8 (R)	23.6 (R)	29.2 (R)	0	refusal
	en	0	9.2 (R)	33.3 (R)	27.8 (R)	0	
Qwen3-ASR	zh/en	0	0	0	0	0	none
Qwen3.8-Omni	zh/en	0	0	n/a	n/a	0	none